

# Boost the Inference with Co-training: A Depth-guided Mutual Learning Framework for Semi-supervised Medical Polyp Segmentation

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## Abstract

*Semi-supervised polyp segmentation has made significant progress in recent years as a potential solution for computer-assisted treatment. Since depth images can provide extra information other than RGB images to help segment these problematic areas, depth-assisted polyp segmentation has gained much attention. However, the utilization of depth information is still worth studying. The existing RGB-D segmentation methods rely on depth data in the inference stage, limiting their clinical applications. To tackle this problem, we propose a semi-supervised polyp segmentation framework based on the mean teacher architecture. We establish an auxiliary student network with depth images as input in the training stage, and we propose a depth-guided cross-modal mutual learning strategy to promote the learning of complementary information between different student networks. Meanwhile, we use the high-confidence pseudo-labels generated by the auxiliary student network to guide the learning progress of the main student network from different perspectives. Our model does not need depth data in the inference phase. In addition, we introduce a depth-guided patch augmentation method to improve the model's learning performance in difficult regions of unlabeled polyp images. Experimental results show that our method achieves state-of-the-art performance under different label conditions on five polyp datasets. The code is available at <https://github.com/pingchuan/RD-Net>.*

## 1. Introduction

Colorectal cancer (CRC) is a common gastrointestinal malignancy and has become the third most common cancer [20]. Various screening methods (such as colonoscopy) are proposed for early detection and treatment to reduce its incidence [18]. Intestinal polyps are often considered a precursor to colorectal cancer [25]. Therefore, the pre-

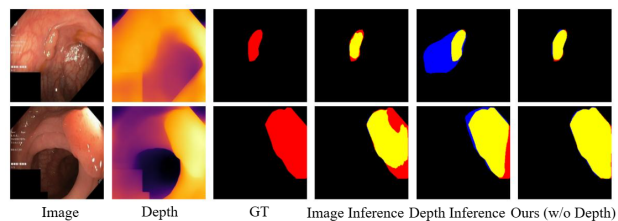


Figure 1. To illustrate the complementary role of RGB and depth images, we present segmentation results on unlabeled images trained with 10% labeled data in the Kvasir-SEG dataset. The results indicate that our method effectively learns richer information from unlabeled data. The red, blue, and yellow areas represent the ground truth, prediction, and their overlap, respectively.

cise segmentation of polyps plays a key role in the early diagnosis and treatment of colorectal cancer. Recently, supervised deep learning methods have achieved remarkable success in the field of polyp segmentation [9, 23, 24, 29, 33]. However, these methods rely on high-quality pixel-wise labeled data. Collecting sufficient labeled data in the medical field is not always feasible. To alleviate this limitation, some studies have explored semi-supervised learning (SSL) [3, 16, 34, 36, 41] methods for medical image segmentation, using unlabeled data to improve the segmentation performance of the model. These semi-supervised learning (SSL) methods mainly rely on RGB images for training, neglecting the rich spatial information in the image. Studies have shown that combining RGB images with depth data can efficiently complement the contour information that can not be easily extracted in RGB images [6, 11, 14, 40]. As shown in Figure 1, the pseudo-labels predicted by the network based on depth images and RGB images can complement each other during training. However, these RGB-D segmentation methods usually require complex feature splicing and fusion in the training phase. This factor leads to reliance on depth data input during the inference phase, greatly limiting their application in clinical practice. To this end, we propose a semi-

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supervised polyp segmentation framework RD-Net based on mean teacher, which effectively takes advantage of RGB and depth images during training and only needs the RGB images without depth information during inference. Specifically, in the training phase, we introduce an auxiliary student network with depth images as input into the traditional teacher-student network framework and adopt the proposed depth-guided cross-modal mutual learning strategy to effectively promote the learning of complementary information between different student networks. Our model uses the high-confidence pseudo-labels generated by the auxiliary student network to guide the training of the main student network from different perspectives. In addition, we propose depth-guided patch augmentation to improve the model’s ability to learn difficult areas of polyp images. We summarize our main contributions as follows:

- A novel semi-supervised polyp segmentation framework is proposed to promote the learning of complementary information between student networks through a depth-guided cross-modal mutual learning strategy. No depth information is required in the inference stage.
- The pseudo-labels generated by the auxiliary student network are additionally used to guide the learning of the main student network from different perspectives, making the most of pixel information.
- Depth-guided patch augmentation is proposed to boost model learning on challenging regions in unlabeled data. Our method outperforms state-of-the-art methods on five challenging polyp datasets.

## 2. Related Work

### 2.1. RGB-D semantic segmentation

RGB-D information contains RGB features such as color texture and depth features such as distance boundaries [8], which are complementary to some extent. Some studies have explored how to integrate RGB-D information [1, 8, 11, 14, 39] to improve the performance of semantic segmentation. ACNet [14] effectively improved the semantic segmentation performance by leveraging the proposed attention complementary modality and using the channel attention mechanism to select bimodal features for fusion at multiple stages. CMX [1] achieved the calibration of different modal features by utilizing the designed cross-modal feature correction module to correct the features of another modality using the features of one modality and effectively fuses the features of different modalities through the proposed feature fusion module. However, these methods often perform complex feature splicing and fusion in the encoder stage. They still rely on depth images in the inference stage, greatly limiting their clinical applications. Although MaskMentor [43] adopted self-training token-pixel joint reconstruction to narrow the intermediate representation between

the missing modality and the complete modality, thereby enhancing the model’s ability to handle the modality missing to some extent. The performance of MaskMentor would decrease if modality was missing. Instead, our method can effectively learn the information between different modalities during the training phase without involving feature fusion between different modalities. Notably, our model does not require depth information during the inference phase.

### 2.2. Semi-supervised medical segmentation

In recent years, a large number of excellent methods have emerged in the field of semi-supervised medical image segmentation [3, 12, 16, 32, 34, 41, 44]. SCPNet [41] proposed a cross-sample prototype learning method to enhance the diversity of predictions in consistency learning by integrating rich semantic information from multiple inputs. BCP [3] encourages unlabeled data to learn richer language information from labeled data by bidirectionally copying and pasting labeled and unlabeled data. ACL-Net [34] proposed a semi-supervised segmentation framework that combines affinity contrast learning and self-training learning, which can adaptively refine pseudo-labels with optimized affinity to improve the performance of semi-supervised polyp segmentation. PH-Net [16] evaluated difficult regions by calculating the hardness scores of image patches and guides the model’s attention to difficult regions, thereby improving segmentation performance. However, this method used the model’s prediction results to identify difficult regions, which are susceptible to network performance. In contrast, we propose a depth-guided patch enhancement that does not rely on the network and can accurately identify difficult regions, thereby strengthening the learning of difficult regions. In addition, we introduce an auxiliary student network trained with depth images and promote the learning of complementary information between different modal networks through a deep-guided cross-modal mutual learning strategy. At the same time, the pseudo-labels of the auxiliary student network are used to adaptively guide the main student network, thereby utilizing more pixel information.

## 3. Method

### 3.1. Preliminary

In semi-supervised polyp segmentation, the dataset consists of two parts: labeled images and unlabeled images. For the labeled images  $\mathcal{D}_l = \{(X_i, Z_i, Y_i)\}_{i=1}^N$ , the unlabeled images  $\mathcal{D}_u = \{X_i, Z_i\}_{i=N+1}^{N+M}$ , where  $X_i \in \mathbb{R}^{H_i \times W_i \times 3}$  is the image,  $Z_i \in \mathbb{R}^{H_i \times W_i \times 3}$  is the corresponding depth image, and  $Y_i \in \{0, 1\}^{H_i \times W_i \times 1}$  represents the ground truth. For the images  $X_i$ , we use a monocular depth estimation network to infer grayscale depth maps of the same size as the original images. Specifically, we use the Depth Anything model [37] and its pre-trained model Depth-Anything-



mutual learning in three aspects based on the consistency and difference of Model\_R1 and Model\_D predictions.

**Consistency learning.** For unlabeled data  $\{X_i, Z_i\} \in \mathcal{D}_u$ , we use consistency constraints to encourage the student network to have consistent prediction. In addition, we also encourage the deep encoding features of the student network outputs of different modalities to be aligned. We use the mean square error (MSE) loss as a constraint for simplicity. The unlabeled data consistency loss can be expressed as:

$$\mathcal{L}_{\text{con}} = \frac{1}{2|\mathcal{D}_u|} \sum_{(X_i, Z_i) \in \mathcal{D}_u} (\|F_i^{r1} - F_i^d\|_2^2 + \|\mathcal{F}_{\theta_{r1}}(X_i^{S1}) - \mathcal{F}_{\theta_d}(Z_i)\|_2^2) \quad (5)$$

where  $F_i^{r1}$  and  $F_i^d$  represent the output features of the last layer encoders of the two student networks, respectively, and  $X_i^{S1} = \text{Aug}_{S1}(X_i)$ .

**Complementary knowledge learning.** By only constraining the consistency of outputs and features of different student networks, this method easily leads to the premature convergence of the two student networks to consistency, making it difficult for the student networks to learn richer cross-modal complementary information. At present, contrastive learning [2, 5, 31, 34] has shown great potential in semi-supervised methods. Considering the large differences between RGB and depth modalities in the feature space, it may be difficult to capture the cross-modal association between them using feature-level contrastive learning directly. Therefore, we use a contrastive learning strategy based on output-level positive and negative sample pairs to guide the model to learn richer cross-modal complementary information. Considering that medical polyp image segmentation is a two-class task, we regard the different category regions of each sample as the same object and establish positive and negative sample pairs between different modalities and different samples. For each batch of training sample set  $\{(X_i, Z_i) \in \mathcal{D}_l\}_{i=1}^K \cup \{(X_i, Z_i) \in \mathcal{D}_u\}_{i=K+1}^{2K}$ , the network output of the paired modality is used as a pair of positive sample pairs. For the positive sample pair set  $T_l^+$  with labeled data  $\{(X_i, Z_i) \in \mathcal{D}_l\}_{i=1}^K$ , it can be formulated as:

$$T_l^+ = \{(\mathcal{F}_{\theta_{r1}}(X_i), \mathcal{F}_{\theta_d}(Z_i))\}_{i=1}^K \quad (6)$$

For the unlabeled data  $\{(X_i, Z_i) \in \mathcal{D}_u\}_{i=K+1}^{2K}$ , the corresponding positive sample pair set  $T_u^+$  can be expressed as:

$$T_u^+ = \{(\mathcal{F}_{\theta_{r1}}(\text{Aug}_{S1}(X_i)), \mathcal{F}_{\theta_d}(Z_i))\}_{i=K+1}^{2K} \quad (7)$$

We can get the positive sample pair set  $\Gamma^+$  under each training batch as:

$$\Gamma^+ = T_l^+ \cup T_u^+ \quad (8)$$

For simplicity, we can express  $\Gamma^+$  as:

$$\Gamma^+ = \{\bar{Y}_i^{r1}, \bar{Y}_i^d\}_{i=1}^{2K} \quad (9)$$

where  $\bar{Y}_i^{r1}$  represents the prediction of the student network Model\_R1, and  $\bar{Y}_i^d$  represents the prediction of the student network Model\_D. For the  $i$ -th positive sample pair, we take the remaining  $4K - 2$  predictions as the  $i$ -th corresponding negative sample pair  $T_i^-$ . The Dice coefficient can measure the overlapping area of two samples and is robust to the imbalance of foreground and background. Inspired by this, we use the Dice coefficient to evaluate the regional similarity of positive and negative samples. The calculation formula for the Dice coefficient can be defined as:

$$S_{\text{Dice}}(\bar{Y}_i^{r1}, \bar{Y}_i^d) = \frac{2 \sum_{k=1}^{H \times W} \bar{Y}_{i,k}^{r1} \cdot \bar{Y}_{i,k}^d}{\sum_{k=1}^{H \times W} \bar{Y}_{i,k}^{r1} + \sum_{k=1}^{H \times W} \bar{Y}_{i,k}^d} \quad (10)$$

where  $\bar{Y}_{i,k}^{r1}$  and  $\bar{Y}_{i,k}^d$  represent the predicted values of the  $k$ -th pixel  $\bar{Y}_i^{r1}$  and  $\bar{Y}_i^d$  predicted by the network respectively.  $H$  and  $W$  represent the predicted height and width respectively. Since the prediction results with RGB images as the main modality have a higher confidence level, when calculating the similarity between the positive sample pairs and the corresponding negative sample pairs, only the prediction  $\bar{Y}_i^{r1}$  obtained by the Model\_R1 is used. Then the contrastive learning loss function  $\mathcal{L}_{\text{cont}}$  can be expressed as:

$$\mathcal{L}_{\text{cont}} = -\frac{1}{|\Gamma^+|} \sum_{(\bar{Y}_i^{r1}, \bar{Y}_i^d) \in \Gamma^+} \log \frac{e^{S_{\text{Dice}}(\bar{Y}_i^{r1}, \bar{Y}_i^d)}}{e^{S_{\text{Dice}}(\bar{Y}_i^{r1}, \bar{Y}_i^d)} + \sum_{\bar{Z}_i^j \in T_i^-} e^{S_{\text{Dice}}(\bar{Y}_i^{r1}, \bar{Z}_i^j)}} \quad (11)$$

where  $\bar{Z}_i^j$  represents the  $j$ -th sample in the negative sample pair corresponding to the  $i$ -th positive sample pair.

**Auxiliary student network guides learning.** We believe that the high confidence prediction of Model\_D on unlabeled data can also be used as a pseudo-label to guide the learning of Model\_R1. For the  $k$ -th pixel prediction  $y_k$ , if  $y_k > 0.5$ , it is a polyp category; if  $y_k \leq 0.5$ , it is a background category. Specifically, we consider the  $k$ -th pixel prediction  $y_k$  of Model\_R2 and the corresponding prediction  $y'_k$  of Model\_D. If the distance between  $y'_k$  and the corresponding category is lower than the threshold distance, and this distance is lower than the distance between  $y_k$  and the corresponding category, we replace the  $k$ -th pixel prediction value with  $y'_k$ , which can be formulated as follows:

$$Y''_{i,k} = \begin{cases} y'_k, & \text{if } y'_k > \gamma \text{ and } d'_k < d_k \\ y'_k, & \text{if } y'_k < 1 - \gamma \text{ and } d'_k < d_k \\ y_k, & \text{otherwise} \end{cases} \quad (12)$$

where  $Y''_{i,k}$  represents the pseudo label of the  $k$ -th pixel of the  $i$ -th image after update. If  $y$  is a polyp category, then



the distance  $d = 1 - y$ ; otherwise,  $d = y$ . Then the corresponding unsupervised loss  $\mathcal{L}_{ud}$  can be formulated as:

$$\mathcal{L}_{ud} = \mathcal{L}_u(\mathcal{F}_{\theta_{r1}}(X_i^{S1}), Y_i'') \quad (13)$$

### 3.3. Depth-guided patch augmentation

The CutMix [38] method has proven its effectiveness [10] in semi-supervised segmentation tasks, and many CutMix-based improvement methods have emerged [16, 17, 42]. PH-Net [16] considers the difference in learning difficulty between difficult and simple regions in an image and proposes an adaptive patch augmentation (APA) method to identify and process difficult regions to improve performance. This method evaluates the difficulty of a patch by calculating the entropy of the model's prediction distribution. However, there are some limitations in evaluating the difficulty of image patches based on model predictions. The model prediction confidence is low when the amount of labeled data is small. Suppose a high-confidence erroneous prediction is made for difficult regions. In that case, these regions may be misjudged as easy regions, thus affecting the overall performance. Instead, we use depth information for difficult patch evaluation. For the RGB image  $X$  and its corresponding single-channel depth map  $D$ , the hardness score of a patch  $P$  of size  $h \times w$  can be expressed as:

$$H_P = \frac{1}{h \times w} \sum_{(i,j) \in P} \text{Var}(D_{i,j}) \quad (14)$$

where  $H_P$  is the hardness score of patch  $P$ ,  $D_{i,j}$  is the depth value of pixel position  $(i, j)$  in the depth map  $D$ , and  $\text{Var}(D_{i,j})$  represents the variance of pixel depth values in the patch. The larger the variance of the patch, the more obvious the depth change of the patch is, which means that the difficulty of the patch is higher. To enhance the richness of the training samples, we randomly select  $h$  and  $w$  from  $\{10, 20, 40\}$ . The patches are sorted in descending order of hardness scores so that patches with higher hardness scores can be shielded to prevent them from being cut off during the cutting process. The number of shielded patches is dynamically set following PH-Net [16]. Then, the depth-guided patch augmentation can be expressed as:

$$\text{Aug}_{S2}(X_i^{S1}) \leftarrow M \odot X_i^{S1} + (1 - M) \odot X_{i+1}^{S1} \quad (15)$$

$$Y_i^{r2} \leftarrow M \odot \tilde{Y}_i^{r2} + (1 - M) \odot \tilde{Y}_{i+1}^{r2} \quad (16)$$

where  $X_i \in \mathcal{D}_u$  and  $X_{i+1} \in \mathcal{D}_u$  represent the  $i$ -th and  $(i + 1)$ -th unlabeled RGB images in the batch.  $\tilde{Y}_i^{r2}$  and  $\tilde{Y}_{i+1}^{r2}$  denote the corresponding pseudo labels predicted by the teacher network for  $X_i$  and  $X_{i+1}$ , and  $M$  is a patch mask randomly selected from the unmasked patches. Then, the corresponding loss  $\mathcal{L}_{u-dpa}$  can be formulated as:

$$\mathcal{L}_{u-dpa} = \mathcal{L}_u(\mathcal{F}_{\theta_{r1}}(\text{Aug}_{S2}(X_i^{S1})), Y_i^{r2}) \quad (17)$$

### 3.4. Loss function

The training loss of student networks consists of supervised loss and unsupervised loss. The training loss of the student network Model\_R1 can be formulated as:

$$\mathcal{L}_{rgb} = \frac{1}{3} (\mathcal{L}_{sup}(\mathcal{F}_{\theta_{r1}}(X_i), Y_i) + \mathcal{L}_{u-dpa} + \mathcal{L}_{ud}) \quad (18)$$

where  $X_i \in \mathcal{D}_l$ ,  $Y_i$  is the ground truth corresponding to the image  $X_i$ . The student network Model\_D predicts  $\tilde{Y}_i^{d,u} = \mathcal{F}_{\theta_d}(Z_i)$  for the unlabeled data  $Z_i \in \mathcal{D}_u$ , while the pseudo label predicted by Model\_R2 using the corresponding RGB image  $X_i$  is  $\tilde{Y}_i^{r2}$ . The training loss of the student network Model\_D can be formulated as:

$$\mathcal{L}_{depth} = \frac{1}{2} (\mathcal{L}_{sup}(\mathcal{F}_{\theta_d}(Z_i), Y_i) + \mathcal{L}_u(\tilde{Y}_i^{d,u}, \tilde{Y}_i^{r2})) \quad (19)$$

where  $Z_i \in \mathcal{D}_l$ ,  $Y_i$  is the corresponding ground truth. The overall training loss can be formulated as:

$$\mathcal{L}_{total} = \mathcal{L}_{rgb} + \mathcal{L}_{depth} + \mathcal{L}_{cont} + \mathcal{L}_{con} \quad (20)$$

## 4. Experiments

### 4.1. Dataset and evaluation metrics

**Datasets.** We evaluated our method on five typical polyp segmentation datasets: Kvasir-SEG [15], CVC-ClinicDB [4], CVC-300 [30], CVC-ColonDB [27] and ETIS-Larib [26].

The Kvasir-SEG dataset contains 1000 polyp images, of which 900 are randomly divided into training sets and the remaining 100 into test sets. The training sets are partitioned into labeled and unlabeled subsets according to the 1/40 and 1/10 partitioning protocols. The CVC-ClinicDB dataset contains 612 polyp images, of which 550 are randomly divided into training sets and the remaining 62 into test sets. The training sets are partitioned into labeled and unlabeled subsets according to the 1/40 and 1/10 partitioning protocols. In addition, we followed the experimental settings of the ACL-Net [34] method. We combined 900 training images from the Kvasir-SEG dataset and 550 training images from the CVC-ClinicDB dataset into a new training set of 1450 images as a semi-supervised training dataset. The training sets are partitioned into labeled and unlabeled subsets according to the 1/10 partitioning protocols. Under this experimental setting, only the CVC-300, CVC-ColonDB, and ETIS-Larib datasets are tested to further evaluate the generalization ability of the model.

**Evaluation metrics.** We adopt three widely used metrics for quantitative evaluation, including the mean dice similarity coefficient (Dice(%)), the mean pixel-wise accuracy (Acc(%)), and the mean intersection over union (IoU(%)).

Table 1. Quantitative comparison using different state-of-the-art methods on Kvasir-SEG and CVC-ClinicDB datasets. Boldface and underlined numbers represent the best and second-best results of each setting, respectively.

| Method       | Kvasir-SEG |            |              |              |              | CVC-ClinicDB |            |              |              |              |
|--------------|------------|------------|--------------|--------------|--------------|--------------|------------|--------------|--------------|--------------|
|              | Labeled    | Unlabeled  | Dice(%)↑     | IoU(%)↑      | Acc(%)↑      | Labeled      | Unlabeled  | Dice(%)↑     | IoU(%)↑      | Acc(%)↑      |
| U-Net        | 22(2.5%)   | 0          | 76.49        | 70.48        | 94.61        | 14(2.5%)     | 0          | 67.60        | 58.45        | 95.90        |
| U-Net        | 90(10%)    | 0          | 82.76        | 77.09        | 95.23        | 54(10%)      | 0          | 77.84        | 72.14        | 97.24        |
| U-Net        | 900(100%)  | 0          | <b>89.43</b> | <b>84.35</b> | <b>97.07</b> | 550(100%)    | 0          | <b>88.34</b> | <b>83.54</b> | <b>98.59</b> |
| MT [28]      | 22(2.5%)   | 878(97.5%) | 64.27        | 67.77        | 93.86        | 14(2.5%)     | 536(97.5%) | 53.19        | 58.79        | 95.67        |
| CCT [21]     |            |            | 44.08        | 44.83        | 84.65        |              |            | 31.02        | 37.14        | 84.43        |
| DTC [19]     |            |            | 73.93        | 70.99        | 94.38        |              |            | 63.25        | 61.76        | 95.76        |
| SCP-Net [41] |            |            | 73.53        | 62.87        | 92.99        |              |            | 63.93        | 53.03        | 94.16        |
| BCP [3]      |            |            | 82.42        | 75.19        | 95.39        |              |            | 76.65        | <u>71.90</u> | 96.75        |
| PH-Net [16]  |            |            | 83.12        | 75.69        | <u>95.65</u> |              |            | 73.37        | 67.42        | 96.17        |
| CML [35]     |            |            | <u>83.21</u> | <u>76.59</u> | 95.43        |              |            | <u>77.60</u> | 71.67        | <u>97.02</u> |
| AD-MT [44]   |            |            | 82.29        | 74.51        | 95.22        |              |            | 74.36        | 66.14        | 96.07        |
| Ours         |            |            | <b>86.35</b> | <b>80.10</b> | <b>96.05</b> |              |            | <b>82.60</b> | <b>75.81</b> | <b>97.32</b> |
| MT [28]      | 90(10%)    | 810(90%)   | 84.81        | 79.27        | 96.13        | 54(10%)      | 496(90%)   | 80.66        | 74.13        | 97.51        |
| CCT [21]     |            |            | 70.02        | 68.39        | 93.09        |              |            | 65.43        | 61.89        | 94.63        |
| DTC [19]     |            |            | 83.84        | 79.74        | 96.11        |              |            | 79.32        | 76.52        | 97.86        |
| SCP-Net [41] |            |            | 85.70        | 78.46        | 96.33        |              |            | 82.08        | 74.93        | 97.70        |
| BCP [3]      |            |            | 87.30        | <u>80.92</u> | 96.25        |              |            | <u>85.59</u> | <u>78.70</u> | <u>97.99</u> |
| PH-Net [16]  |            |            | <u>87.35</u> | 80.67        | <u>96.54</u> |              |            | 82.11        | 75.06        | 97.93        |
| CML [35]     |            |            | 86.89        | 80.46        | 95.72        |              |            | 85.49        | 77.92        | 97.97        |
| AD-MT [44]   |            |            | 86.38        | 79.91        | 96.23        |              |            | 83.66        | 76.22        | 97.42        |
| Ours         |            |            | <b>89.07</b> | <b>84.21</b> | <b>97.07</b> |              |            | <b>87.84</b> | <b>81.84</b> | <b>98.15</b> |

## 4.2. Implementation details

All experiments in this study were implemented on a single NVIDIA GeForce RTX 3060 using the PyTorch framework and used a fixed random seed. Our baseline uses the UNet [22] network and uses the ResNet34 [13] pre-trained on ImageNet [7] as the UNet backbone. The image and label sizes are resized to  $320 \times 320$ . The SGD optimizer is used with an initial learning rate of 0.001, momentum of 0.9, and weight decay of 0.00001. The learning rate is decayed using a polynomial strategy:  $\eta = \eta_{\text{init}} \cdot \left(1 - \frac{\text{iter}}{\text{iter}_{\text{total}}}\right)^{0.9}$ . In our experimental setup, we adopt a batch size of 4 and train for 50,000 iterations, with the batch size for labeled and unlabeled data set to 2.

## 4.3. Comparisons with SOTA methods

We compare our method with eight state-of-the-art semi-supervised segmentation methods, including MT [28], CCT [21], DTC [19], SCP-Net [41], BCP [3], PH-Net [16], CML [35], and AD-MT [44]. To ensure a fair comparison, for all experiments, we use UNet with ResNet-34 as the backbone of the segmentation network, adopt the same experimental environment and data augmentation, and the data partitioning protocol of these methods is the same.

As shown in Table 1, our proposed method achieves the highest scores on the three evaluation metrics on two typical polyp segmentation datasets, Kvasir-SEG and CVC-ClinicDB, with 2.5% and 10% labeled data settings. On the Kvasir-SEG dataset, using only 22 labeled images, our method achieves a Dice score of 86.35% and an IoU

score of 80.10%, which is 3.14% higher in Dice score and 3.51% higher in IoU score than the previous state-of-the-art method CML [35]. To more intuitively compare the performance of semi-supervised methods, we also conducted experiments in a fully supervised manner. Under the 10% experimental setting, our method approaches the performance of full supervision. Our method achieves the best segmentation under all partitions of the Kvasir-SEG dataset and CVC-ClinicDB dataset, indicating that our method enables the model to be better trained on unlabeled images.

We also experimentally demonstrate that our method has excellent generalization ability on the unseen CVC-300, CVC-ColonDB, and ETIS-Larib polyp datasets. As shown in Table 2, our method has surpassed competitors significantly. Specifically, compared with the SupOnly method, our method achieves a 10.44% Dice and 8.80% IoU score improvement on the CVC-ColonDB dataset. We also obtained a 28.17% Dice and 23.67% IoU score improvement on the most challenging ETIS-Larib dataset. These experimental results show that by combining the mutual learning strategy of depth images and RGB images, our method can significantly enhance the model’s feature representation ability. This approach helps the model better understand the shape and structure of polyps, which is particularly important for delicate tasks such as segmenting polyps. Therefore, our model performs well on different datasets, showing its strong adaptability and effectiveness.

For qualitative analysis, to more intuitively demonstrate our method’s superiority, we visually compare challenging cases in Figure 3. Our method produces excellent segmen-

Table 2. Generalization ability comparison with different state-of-the-art methods. We split all the training images (1450) in Kvasir-SEG and CVC-ClinicDB into 1/10 labeled images (144) and 9/10 unlabeled images (1306) for training. “SupOnly” stands for supervised training without using any unlabeled data. Boldface and underlined numbers represent the best and second best results of each setting, respectively.

| Method       | Kvasir-SEG         |                   | CVC-ClinicDB       |                   | CVC-300            |                   | ETIS-Larib         |                   | CVC-ColonDB        |                   |
|--------------|--------------------|-------------------|--------------------|-------------------|--------------------|-------------------|--------------------|-------------------|--------------------|-------------------|
|              | Dice(%) $\uparrow$ | IoU(%) $\uparrow$ | Dice(%) $\uparrow$ | IoU(%) $\uparrow$ | Dice(%) $\uparrow$ | IoU(%) $\uparrow$ | Dice(%) $\uparrow$ | IoU(%) $\uparrow$ | Dice(%) $\uparrow$ | IoU(%) $\uparrow$ |
| SupOnly      | 82.22              | 77.33             | 74.71              | 69.39             | 79.10              | 70.47             | 43.30              | 39.80             | 63.85              | 57.55             |
| MT [28]      | 84.60              | 78.52             | 76.43              | 69.57             | 76.84              | 70.12             | 48.45              | 42.17             | 69.23              | 62.41             |
| CCT [21]     | 84.93              | 79.63             | 76.68              | 70.20             | 81.72              | 75.66             | 52.04              | 47.63             | 66.14              | 59.43             |
| DTC [19]     | 84.61              | 79.75             | 77.02              | 73.60             | 81.16              | 75.22             | 52.89              | 49.71             | 65.67              | 61.49             |
| SCP-Net [41] | 85.45              | 77.75             | 79.79              | 72.95             | 79.79              | 71.50             | 48.57              | 40.35             | 66.10              | 57.57             |
| BCP [3]      | 86.25              | <u>80.10</u>      | 81.75              | 75.09             | 80.36              | 76.12             | 59.70              | 51.72             | 66.19              | 58.61             |
| PH-Net [16]  | <u>87.02</u>       | 79.95             | 79.01              | 72.64             | 84.35              | 77.24             | 59.73              | 51.85             | 69.70              | 61.62             |
| CML [35]     | 85.39              | 79.01             | <u>84.01</u>       | <u>77.77</u>      | 83.10              | 75.98             | 56.00              | 49.45             | 63.88              | 57.51             |
| AD-MT [44]   | 85.55              | 78.28             | 82.13              | 76.07             | <u>84.43</u>       | <u>77.52</u>      | <u>66.29</u>       | <u>58.61</u>      | <u>71.46</u>       | <u>64.13</u>      |
| Ours         | <b>88.88</b>       | <b>82.98</b>      | <b>85.88</b>       | <b>79.97</b>      | <b>85.61</b>       | <b>79.32</b>      | <b>71.47</b>       | <b>63.47</b>      | <b>74.29</b>       | <b>66.35</b>      |

tation results for very small polyps with unclear boundaries and low contrast areas between polyps and their surrounding mucosa. This indicates that our method can effectively deal with challenges such as polyp size, shape, and low contrast.

Table 3. Ablation study of different components on Kvasir-SEG dataset with 10% labeled data. DPA: Depth-guided patch augmentation. “SupOnly” stands for training using only labeled data.

| Method  | $\mathcal{L}_{ud}$ | $\mathcal{L}_{con}$ | $\mathcal{L}_{cont}$ | DPA | Dice (%)     | IoU (%)      |
|---------|--------------------|---------------------|----------------------|-----|--------------|--------------|
| SupOnly |                    |                     |                      |     | 82.76        | 77.09        |
| I       | ✓                  | ✓                   |                      |     | 87.96        | 82.61        |
| II      | ✓                  |                     | ✓                    |     | 88.08        | 82.51        |
| III     | ✓                  |                     |                      | ✓   | 88.03        | 82.68        |
| IV      | ✓                  | ✓                   | ✓                    |     | 88.14        | 82.89        |
| V       | ✓                  | ✓                   | ✓                    | ✓   | <b>89.07</b> | <b>84.21</b> |

Table 4. Ablation study of different enhancement methods on Kvasir-SEG dataset. SA: Aug<sub>S1</sub>.

| Method         | # images used |            | Metrics      |              |
|----------------|---------------|------------|--------------|--------------|
|                | Labeled       | Unlabeled  | Dice (%)     | IoU (%)      |
| SA             |               |            | 81.67        | 74.79        |
| SA+CutMix [38] | 22(2.5%)      | 878(97.5%) | 84.53        | 78.44        |
| SA+APA [16]    |               |            | 84.42        | 77.87        |
| SA+DPA         |               |            | <b>86.35</b> | <b>80.10</b> |
| SA             |               |            | 87.16        | 81.01        |
| SA+CutMix [38] | 90(10%)       | 810(90%)   | 88.09        | 82.27        |
| SA+APA [16]    |               |            | 88.50        | 82.76        |
| SA+DPA         |               |            | <b>89.07</b> | <b>84.21</b> |

Table 5. Ablation study of confidence threshold  $\gamma$  on Kvasir-SEG dataset with 10% labeled data.

| $\gamma$ | 0.7   | 0.75  | 0.8   | 0.85         | 0.9   | 0.95  |
|----------|-------|-------|-------|--------------|-------|-------|
| Dice (%) | 88.66 | 88.74 | 89.06 | <b>89.07</b> | 89.03 | 88.72 |
| IoU (%)  | 83.60 | 83.55 | 83.91 | <b>84.21</b> | 83.74 | 83.67 |

#### 4.4. Ablation studies

**Effectiveness of components.** In Table 3, we study the effectiveness of our proposed RD-Net main components on the Kvasir-SEG dataset with 10% labeled data. We use the model trained with supervised learning (SupOnly) using only labeled data as our baseline. The model with depth-guided cross-modal mutual learning ( $\mathcal{L}_{ud}$ ,  $\mathcal{L}_{con}$  and  $\mathcal{L}_{cont}$ ) can obtain a performance gain of the Dice of the baseline by +5.38%, verifying that depth-guided cross-modal mutual learning can provide more cross-membrane complementary information for model training. Adding Depth-guided patch augmentation (DPA) improves the Dice of the baseline by +5.27%, verifying that the DPA module can guide the model to learn difficult patch regions. The full model with all components obtains the most accurate segmentation with the highest Dice and IoU scores. Overall, the results demonstrate the importance of each component in our proposed method for semi-supervised polyp segmentation. To more clearly demonstrate the advantages of each component, Figure 4 shows the predicted visualization of our method. Obviously, with the gradual introduction of enhanced components, the predicted lesion boundaries gradually become more refined. This result shows the effectiveness of our method in processing complex polyp images.

**Effectiveness of depth-guided cross-modal mutual learning.** We evaluated the prediction accuracy of the unlabeled data in the training set, and the results are shown in Figure 5. Notably, by adopting the depth-guided cross-modal mutual learning strategy, the model can improve the segmentation accuracy of the unlabeled training data in each partition of the Kvasir-SEG and CVC-ClinicDB datasets. This result shows that our depth-guided cross-modal mutual learning strategy effectively helps the model better utilize the information of unlabeled images.

**Effectiveness of Depth-guided patch augmentation.** As shown in Table 4, we compare the performance of APA augmentation, ordinary CutMix augmentation, and our DPA

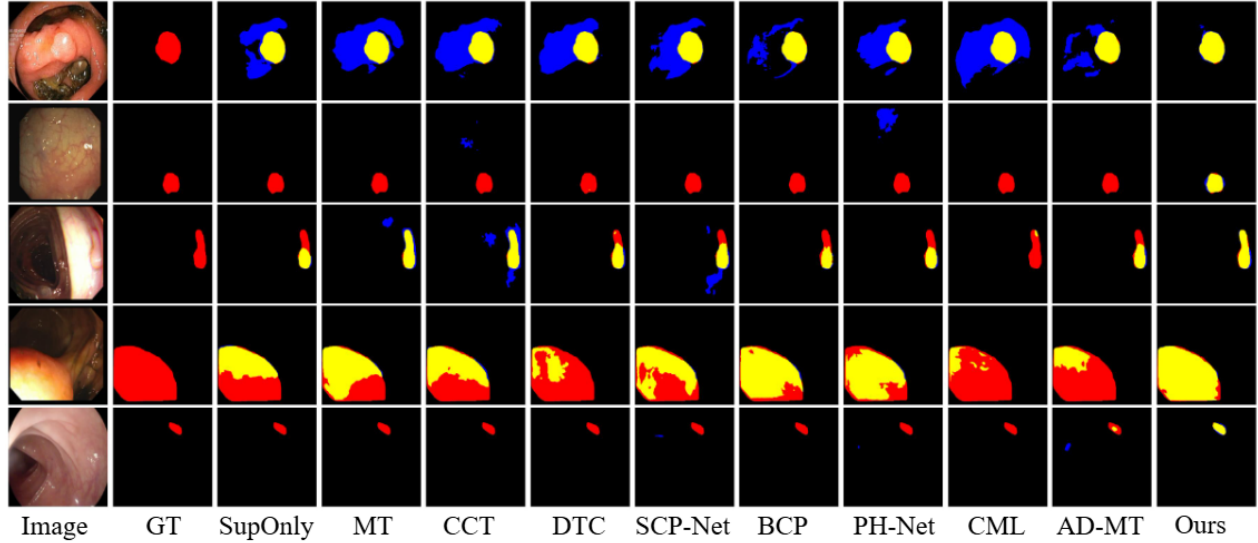


Figure 3. Visual comparison with different state-of-the-art methods on five public polyp datasets. The SupOnly method is trained on only 1/10 labeled data. The red, blue, and yellow areas represent the ground truth, predictions, and their overlap regions, respectively.

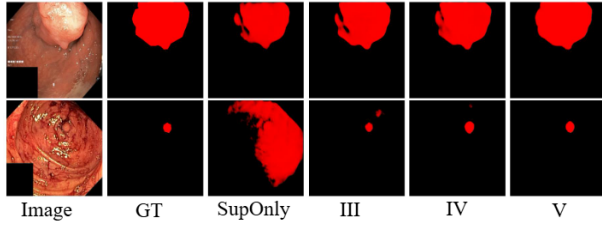


Figure 4. Visual comparison of component ablation studies. Darker red indicates higher confidence in the prediction.

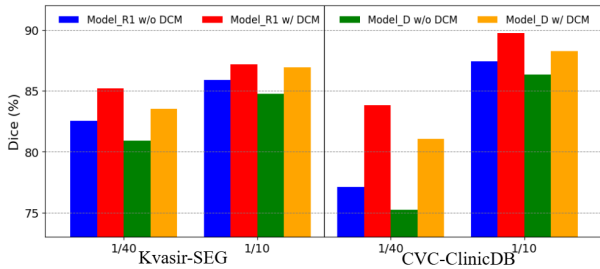


Figure 5. Comparing predictions on unlabeled training data under all partitioning protocols. DCM: Depth-guided cross-modal mutual learning.

augmentation. When the labeling ratio is 2.5%, the Dice value of DPA augmentation is 1.93% higher than that of APA augmentation, which indicates that in the case of very little labeled data, the prediction confidence of the model is low, resulting in poor performance of APA based on prediction for masking patches. In contrast, our method can more effectively select difficult patches for masking, thereby effectively identifying and alleviating the problem of insufficient training of challenging patches.

**Ablation study on hyper-parameters.** Table 5 shows the ablation study of the confidence threshold  $\gamma$  used to filter pseudo labels in Eq.(3) on the Kvasir-SEG dataset. As the threshold increases from 0.7 to 0.95, the Dice and IoU scores of our method increase first and then decrease. This result indicates that the model is prone to learning wrong prediction information when the threshold is low. Additionally, too much correct prediction information may be filtered out when the threshold is too high. We set  $\gamma = 0.85$ .

## 5. Conclusion

This paper proposes a novel semi-supervised polyp segmentation framework. The proposed depth-guided cross-modal mutual learning strategy promotes the learning of complementary information between different student networks in the training phase, while no depth information is required in the inference phase. At the same time, the pseudo-labels generated by the auxiliary student network are additionally used to guide the learning of the main student network from different angles, making the most of more high-confidence pixels. In addition, we introduce a depth-guided patch augmentation method to promote the model’s learning in difficult areas of polyp images. Experimental results show that our method significantly outperforms the state-of-the-art methods on five polyp datasets.

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